

## Abirami Valliammai Kathiresan

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### Education:

#### University of California, Davis | Davis, CA

September 2023 - June 2027

- B.S. in Data Science, Minor in Computer Science & Tech Management | GPA: 3.95, Deans' Honor List (Fall 2023 - Winter 2026)
- Relevant Coursework: Data Structures in Python, Data Structures in C++, Databases, Machine Learning, Algorithms for Data Science, Discrete Mathematics, Statistics for Data Analytics, Calculus, Linear Algebra

### Experience:

#### Software Engineering Intern | Charles Schwab

June 2026 - Current

- Migrating 5+ infrastructure automation playbooks with Ansible and YAML for an internal tool monitoring application and service health, streamlining reliability workflows to support scalable operations across Schwab's container ecosystem.
- Developing 10+ REST APIs with Java to support automated diagnostics and health validation for enterprise applications and services.
- Building and deploying AI agents to streamline operational workflows and surface diagnostic insights for the Container Platform Reliability team, leveraging LLM-powered automation to optimize troubleshooting workflows by up to 30%.

#### Software Developer | ASUCD Innovation & Research Lab

January 2026 - Current

- Developing and maintaining the UC Davis mobile application using React Native, TypeScript, Expo, SQLite, and REST APIs in collaboration with the student government, undergoing beta testing before deployment to 40,000+ UC Davis students in Fall 2026.
- Engineered library room reservation system with optimistic locking, while using SQLite and Drizzle to mitigate third-party API delays.
- Debugged and optimized real-time transit prediction workflows across 20+ bus routes, improving arrival-time accuracy and reliability.

#### Software Engineering Intern | Equitas Bank

June 2025 - August 2025

- Developed 7+ REST APIs using Java and Spring Boot to support key internet banking operations, shipping code quickly and achieving average response times below 200 ms while collaborating with cross-functional teams in an Agile development environment.
- Built a backend-for-front-end (BFF) layer to streamline communication between microservices and the frontend, reducing API latency by 30% for thousands of daily users, and deployed services on AWS to ensure scalability in a fast-paced environment.
- Implemented and validated SQL queries on Oracle databases using Postman for 50+ tests, ensuring data accuracy and reliable API performance across backend services and banking applications.

#### Product Management Fellow | Product Space

October 2024 - Current

- Participated in a fellowship program to design and prototype Instagram Scraps, a feature that enabled Instagram users to customize their profiles, showcasing an ability to tackle end-to-end engineering challenges and collaborate cross-functionally.
- Leveraged interdisciplinary skills in product management to design and engineer a user-centric solution and developed interactive wireframes and prototypes in Figma, integrating direct user feedback to enhance interface clarity by 40%.

#### Full-Stack Developer | UC Davis #include

October 2024 - March 2025

- Designed and developed a responsive web application for the Joan Viteri Memorial Clinic using React, Node.js, HTML, CSS, and JavaScript, supporting 1000+ daily active users, ensuring fast iteration cycles and effective API integration.
- Implemented modern web development practices, including dynamic content rendering and responsive design, which increased mobile engagement by 30% and reduced load times by 25% in a collaborative, agile environment.

### Projects:

#### ANCOR

- Developed a full-stack political literacy platform using React, Node.js, Firebase, and web scraping, to aggregate political news, ballot propositions, and educational resources and make government information more accessible to voters.
- Implemented scalable backend services and responsive UI components, enabling real-time delivery of 200+ political terms, legislative updates, and election-related articles across desktop and mobile devices.

#### RiskPredictor

- Designed a machine learning system to help international students assess deportation risk by combining demographic data with trends from 200+ web scraped immigration-related news articles.
- Trained classification models to 92% accuracy with preprocessing and feature selection, and deployed a Streamlit interface for real-time, visualization-supported predictions.

### Technical Skills:

- Languages: Java, Python, JavaScript, TypeScript, HTML, CSS, C++, R, SQL, Bash, YAML
- Frameworks and Libraries: Spring Boot, React, Node.js, Express.js, BrainFlow, MNE, scikit-learn, PyTorch, NumPy, Matplotlib, Spark
- Tools: Git, AWS, Linux, Docker, Gradle, Maven, Jira, Postman, Figma, Streamlit, Oracle, MySQL, PostgreSQL, MongoDB, PCF